Perceptual Losses for Real-Time Style Transfer and Super-Resolution

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The problems related to the task of converting an input image into an output image by transforming the pixels of one into a new form through some model or rule, called image transformation tasks, is mainly focused on in the original paper[6]. Examples of it would include denoising or colorization[1] (image processing) and semantic segmentation or depth estimation [2, 3] (computer vision). For conducting such tasks, most existing methods used feed-forward convolutional neural networks (CNNs) trained with a per-pixel loss. That means the network learns by pixel-by-pixel in the output to the corresponding pixel in the ground truth. But, the problem with this method is that per-pixel losses do not capture how humans actually see images, meaning that even a small misalignment or texture differences can give a high loss, without reflecting how humans perceive image similarity. (e.g. If two images are identical but one is shifted by 1 pixel, the per-pixel loss will still say that they are very different, even though they look almost the same.) It fails to capture perceptual similarity such as texture, structure, or overall look.

To overcome such a problem, recent work has proposed a new direction called "perceptual loss functions." They measure image similarity using high-level features, such as textures, shapes, or patterns, which are extracted from a pretrained CNN such as VGG. This enables the model to compare feature activations, which are the representations of textures, shapes, and semantic content, instead of comparing pixel values. Despite its ability to produce images that look more natural and visually pleasing, the drawback of slow processing speed persists because they require iterative optimization for each image.

The original paper[6] contributed by combining the speed of the aforementioned feed-forward CNNs with the quality of perceptual losses. They focused on training feed-forward transformation networks using perceptual loss instead of per-pixel loss so that it would depend on high-level features from a pretrained loss network. During training, perceptual loss provides a more reliable measure of image similarity than pixel-wise loss, and once trained, the transformation network can generate results instantly in real time. The authors test their method on style transfer and super-resolution, both of which are ill-posed problems requiring the network to understand image semantics. Using perceptual loss lets the model inherit semantic knowledge from a pretrained network instead of learning from scratch. As a result, their feed-forward network achieves style transfer with quality similar to Gatys et al[4]. but about 1000× faster and produces sharper, more natural results for ×4 and ×8 super-resolution.

The proposed system consists of two main components: the *Image Transformation Network* (f_W) and the *Loss Network* (ϕ). The transformation network is a deep residual convolutional neural network that maps an input image x to an output image $\hat{y}$ via $\hat{y} = f_W(x)$. It is trained to minimize a weighted sum of perceptual losses defined by the fixed, pretrained loss network ϕ:

$$W^* = \arg\min_W E_{x,\{y_i\}} \left[\sum_i \lambda_i \ell_i(f_W(x), y_i) \right].$$

The *Loss Network* (ϕ), implemented with a pretrained VGG-16, encodes high-level semantic and visual features used to measure perceptual similarity between images. It defines two key loss terms: the *Feature Reconstruction Loss* (ℓ^{ϕ}_{feat}) for preserving image content, and the *Style Reconstruction Loss* (ℓ^{ϕ}_{style}) for transferring texture and color. For each input image x, there is a content target y_c and a style target y_s: in style transfer, $y_c = x$ and the network learns to produce an output $\hat{y}$ combining the content of x with the style of y_s, while in super-resolution, x is a low-resolution input and y_c is the high-resolution ground truth, using only the content loss.

The *Image Transformation Network* (f_W) follows a residual architecture inspired by Radford et al[9] and He et al[5]. It uses strided convolutions for downsampling and fractionally strided convolutions for upsampling, avoiding pooling layers. The network body contains five residual blocks with batch normalization and ReLU activations, and the output layer uses a scaled tanh to constrain pixel values to $[0, 255]$. The first and last layers use 9×9 kernels and all others use 3×3 kernels. Downsampling and upsampling are learned jointly with the network, improving efficiency and increasing the effective receptive field for coherent global transformations. Residual connections help preserve the spatial structure between input and output images.

In addition to perceptual losses, two simple low-level terms are included: the *Pixel Loss*, which computes the normalized Euclidean distance between $\hat{y}$ and y, and the *Total Variation (TV) Regularization*, which encourages spatial smoothness and reduces artifacts in the generated images.

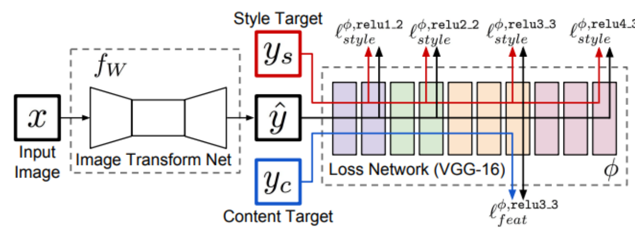

Figure 1: System overview of the proposed method. The image transformation network (f_W) generates the output image $\hat{y}$, while the loss network (ϕ) computes perceptual losses based on high-level features from a pretrained VGG-16 to guide training.

The proposed method is evaluated on two tasks: *style transfer* and *single-image super-resolution*. In both cases, using perceptual loss instead of per-pixel loss produces visually superior results while running up to **1000× faster** than optimization-based methods.

For *style transfer*, one transformation network is trained per artistic style using the MS-COCO dataset[8] (80k images, resized to 256×256). The network uses perceptual losses computed from the VGG-16 layers `relu2_2` for content and `relu1_2`, `relu2_2`, `relu3_3`, `relu4_3` for style. Training uses Adam [7] (learning rate 1×10^{-3}) for 40k iterations and takes about 4 hours on a Titan X GPU. The feed-forward model achieves quality comparable to Gatys et al. [4] but performs inference in real time (20 FPS at 512×512 resolution). The model generalizes to higher resolutions and preserves semantic structures such as faces and objects more effectively than background textures.

For *super-resolution*, networks are trained for ×4 and ×8 upscaling using the feature reconstruction loss at layer `relu2_2`. Models trained with perceptual loss reconstruct sharper edges and finer textures compared to those trained with pixel loss. Evaluated on Set5, Set14, and BSD100 datasets, perceptual-loss models yield slightly lower PSNR but produce more visually realistic and detailed images than baseline methods such as SRCNN. The results show that feature-based losses enable semantic understanding and perceptually pleasing image restoration.

In summary, the proposed approach unites feed-forward efficiency with perceptual quality, achieving real-time style transfer and detailed super-resolution results. Future work includes applying perceptual losses to other image transformation tasks such as colorization and semantic segmentation, and exploring alternative pretrained loss networks to transfer different semantic representations.

[1] Zhicheng Cheng, Qifeng Yang, and Bin Sheng. Deep colorization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 415–423, 2015.

[2] David Eigen and Rob Fergus. Predicting depth, surface normals and semantic labels with a common multi-scale convolutional architecture. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2650–2658, 2015.

[3] David Eigen, Christian Puhrsch, and Rob Fergus. Depth map prediction from a single image using a multi-scale deep network. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 2366–2374, 2014.

[4] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. A neural algorithm of artistic style. *arXiv preprint arXiv:1508.06576*, 2015.

[5] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *arXiv preprint arXiv:1512.03385*, 2015. URL https://arxiv.org/abs/1512.03385.

[6] Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual losses for real-time style transfer and super-resolution. *arXiv preprint arXiv:1603.08155*, 2016. URL https://arxiv.org/abs/1603.08155.

[7] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. URL https://arxiv.org/abs/1412.6980.

[8] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer Vision – ECCV 2014*, pages 740–755. Springer, 2014.

[9] Alec Radford, Luke Metz, and Soumith Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*, 2015. URL https://arxiv.org/abs/1511.06434.